Few-Shot Learning With Mutual Information Enhancement for Hyperspectral Image Classification

Qiaoli Zhang, Jiangtao Peng, *Senior Member, IEEE*, Weiwei Sun, *Senior Member, IEEE*, and Quanyong Liu

***Abstract*— In recent years, few-shot learning (FSL) has made significant progress in hyperspectral image classification (HSIC) by transferring metaknowledge from a labeled source domain to a target domain with very limited labeled samples. Considering that natural images have rich spatial texture information, heterogeneous FSL (HFSL) by using natural images as the source domain and hyperspectral image (HSI) as the target domain has shown excellent performance. However, some problems also exist in the HFSL, such as poor generalization ability from natural images to HSIs, prototype instability due to limited labeled samples, and domain shift between different types of images. To address these problems, we propose a mutual information enhancement FSL (MIEFSL) method for HSIC, which mainly contains three modules, i.e., mutual information enhancement (MIE), intradomain prototype rectification (IPR), and interdomain distribution alignment (IDA). In order to improve the generalization ability of the network and preserve the raw data information as much as possible, an MIE module is designed to maximize the mutual information (MI) between the support set samples and their corresponding masked samples. To stabilize the prototypes, an IPR module is constructed through a distribution expansion strategy. In addition, to alleviate domain shifts between different types of images, an IDA is performed between source and target domains. Experimental results demonstrate that the proposed MIEFSL outperforms existing state-of-the-art FSL methods and achieves the overall accuracy (OA) of 78.34%, 90.31%, and 91.72% on Indian Pines (IP), University of Pavia (UP), and Salinas (SA) in the case of only five labeled samples, respectively.**

***Index Terms*— Few-shot learning (FSL), hyperspectral image classification (HSIC), mutual information (MI), prototype rectification.**

NOMENCLATURE

D_s	Source-domain dataset.
C_s	Source-domain class.
S_s	Source-domain support set.
Q_s	Source-domain query set.
$\boldsymbol{x}_i^{\mathrm{ss}}, y_i^{\mathrm{ss}}$	ith sample and corresponding labels in the source-domain support set.
$\boldsymbol{x}_j^{\mathrm{sq}}, y_j^{\mathrm{sq}}$	jth sample and corresponding labels in the source-domain query set.
$\boldsymbol{A}_c^s$	cth category prototype in the support set of the source domain.
$\widehat{\boldsymbol{A}_c^s}$	cth calibrate category prototype in the support set of the source domain.
$\mathcal{L}_{\mathrm{IPR}}$	Loss of IPR.
$\mathcal{L}_{\mathrm{MIE}}$	Loss of MIE.
$\mathcal{L}_{\mathrm{IDA}}$	Loss of IDA.
$\mathcal{L}_s^{\mathrm{total}}$	Total loss of the source-domain training.
D_t	Target-domain dataset.
C_t	Target-domain classes.
S_t	Target-domain support set.
Q_t	Target-domain query set.
$\boldsymbol{x}_i^{\mathrm{ts}}, y_i^{\mathrm{ts}}$	ith sample and corresponding labels in the target-domain support set.
$\boldsymbol{x}_j^{\mathrm{tq}}, y_j^{\mathrm{tq}}$	jth sample and corresponding labels in the target-domain query set.
$\mathcal{L}_{\mathrm{FSL}}$	Loss of FSL.
$\mathcal{L}_{\mathrm{CL}}$	Loss of CL.
$\mathcal{L}_t^{\mathrm{total}}$	Total loss of target-domain training.

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I. INTRODUCTION

HYPERSPECTRAL image (HSI) is a special type of remote sensing image with high spectral resolution, which can capture spectral information of objects on dozens or even hundreds of continuous spectral bands. The rich spectral information combined with wide-coverage spatial information makes HSIs widely used in Earth observation tasks. HSI classification (HSIC) can enable the accurate identification of ground objects, providing important support for Earth observation tasks such as environmental monitoring, resource management, and urban planning [1]. In the early stage of HSIC, traditional classifiers are usually performed on the original spectral features [2], [3] or handcraft features, such as morphological features [4] and Gabor features [5]. Although traditional HSIC methods have made significant progress in the early stages, their performance largely depends on the effectiveness of the handcraft features.

In recent years, deep learning-based methods have gradually demonstrated their advantages in handling high-dimensional complex data [6], [7]. They can automatically mine task-related features from different types of data and become research hotspots in HSIC [7], [8]. Chen et al. [8] first

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proposed a stacked autoencoder (SAE)-based deep network architecture to extract high-level spatial–spectral features for HSIC. To reduce the network parameters, they further proposed a convolutional neural network (CNN)-based HSIC method [9]. By using local receptive fields and weight sharing strategy, CNN has shown excellent performance for HSIC. Subsequently, many CNN variants have been proposed for HSIC [10], [11], [12]. To solve the gradient explosion or vanishing problem as the increase of network layers, Zhong et al. [13] proposed a spectral–spatial residual network (SSRN). To balance the computational complexity and classification performance, Roy et al. [14] proposed a hybrid spectral CNN (HybridSN), which used 3-D-CNN to extract spectral–spatial joint features followed by 2-D-CNN to further extract spatial features. Although CNN-based methods can automatically extract classification-related discriminative features, they usually require a large number of labeled samples to train the network [15], [16], [17].

As the labeling of land cover pixels is time-consuming and costly, we often face the classification situation with very few labeled samples. When the number of labeled samples is limited, the classical CNN or deep learning methods may be degraded or fail. To solve the small sample deep learning problem, few-shot learning (FSL) was proposed [18], [19]. Inspired by the human vision system's ability to quickly learn new categories with a small number of labeled samples, FSL can make accurate predictions by training on a very small number of labeled examples. FSL adopts episodic training mode and mainly consists of two stages, i.e., metatraining and metatesting [19]. In the metatraining stage, a large number of metatasks are constructed to train models and obtain metaknowledge. In the metatesting stage, the learned metaknowledge is transferred to new unknown tasks. On labeled source data, FSL uses a large number of small sample metatasks to train the network such that it has a small sample classification ability. Then, on target data with limited labeled samples, the trained network can be fine-tuned to provide accurate predictions even in unknown classes.

In the FSL framework, if we have labeled HSI data, then we can perform the small sample classification for other HSIs [20]. Liu et al. [21] proposed a deep FSL (DFSL) framework, which addressed the problem of sample scarcity in HSIC using the FSL framework of a 3-D residual network. Li et al. [22] proposed a deep cross-domain FSL method (DCFSL) for HSIC, which mainly considered the domain shift problem in the FSL framework and introduced a conditional adversarial domain adaptation strategy to solve the problem. Similarly, Zhang et al. [23] adopted a graph information aggregation strategy to alleviate domain shift between source and target data and proposed a graph information aggregation cross-domain FSL (Gia-CFSL) method. Xi et al. [24] proposed a class covariance metric-based FSL (CMFSL) method for HSIC, which assigned class labels directly using the Mahalanobis distance. Liu et al. [25] proposed a refined prototype contrastive learning (CL)-based FSL (RPCL-FSL) method, which integrated the CL and FSL into an end-to-end network for HSIC. Ye et al. [26] proposed an adaptive domain-adversarial FSL (ADAFSL) method, which assigns different weights to the conditional adversarial losses for different categories to achieve global conditional distribution alignment.

The aforementioned FSL methods are homogeneous FSL methods, which use either the single Chikusei hyperspectral dataset or mixed hyperspectral dataset as the source domain. For the Chikusei dataset, the number of land cover categories is only 19. To enlarge the number of categories, a mixed hyperspectral dataset is constructed [21], [24]. It consists of four hyperspectral datasets, i.e., Chikusei, University of Houston 2013, Botswana, and Kennedy Space Center, and has 77 classes. As the four hyperspectral datasets are acquired by different sensors, the mixed data have no specific physical meaning from spectral and spatial aspects. Compared with HSIs, natural images have rich spatial texture information, and the number of labeled categories is huge. Therefore, using natural image datasets as source data is likely to improve the network's spatial feature extraction ability and generalization ability for unknown classes. Wang et al. [27] proposed a heterogeneous FSL (HFSL) framework for HSIC, which performed metaknowledge transferring from natural images to HSIs. Li et al. [28] proposed an FSL framework with self-supervised learning (FSCF-SSL) for HSIC, which designed a geometric transformation module to learn rich spatial information of natural images to improve the performance of HSIC. However, the above two methods did not fully take into account the significant differences between natural images and HSIs. In the metatraining stage on natural images, they use rotation transformations to generate contrastive groups and perform CL to train the VGG16 spatial feature extraction network [28]. The rotation for natural images is meaningful. However, for HSI, the rotation of a spatial patch may be not effective. In addition, the CL maximizing the shared information between two views has the risk of over-fitting [29]. In the metatesting stage, directly transferring the model learned from natural images to HSIs may degrade the model performance due to the domain shift between natural images and HSIs [30].

To address the above problems, we propose a mutual information enhancement-based FSL (MIEFSL) method for HSIC, which uses the natural image dataset as the source domain and the HSI dataset as the target domain. The proposed MIEFSL mainly contains three modules, i.e., mutual information enhancement (MIE), intradomain prototype rectification (IPR), and interdomain distribution alignment (IDA). In the MIE module, a masking strategy is used to generate contrastive groups. Then, the mutual information (MI) between features in two different contrastive groups and MI between feature and input are simultaneously maximized to learn classification-related information. MI can enhance feature representation by maximizing the nonshared information between input data and its output feature, thereby improving the model's generalization ability to new small sample classification tasks and alleviating the overfitting phenomenon [31], [32], [33]. By using MI, more information about the original data can be retained, resulting in more robust and accurate models even when labeled samples are limited. In the IPR module, the support set prototypes are rectified using the cluster centers of the query set, which can alleviate the problem of prototype

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instability due to limited labeled samples. To mitigate the difference between natural images and HSIs, an IDA is performed on the query sets of source and target domains. In summary, the main contributions of this article are given as follows.

1) An HFSL method MIEFSL is proposed for cross-domain HSIC. It fully exploits spatial texture information and rich category information of natural images and effectively transfers metaknowledge from natural images to HSIs through three modules, i.e., MIE, IPR, and IDA.
2) An MIE module is designed to maximize the MI between support set features and their corresponding features in the masked support set. In addition, the MI between original/masked support set features and inputs is also maximized to learn nonshared classification-related information, which is likely to improve the network's generalization ability.
3) An IPR module is constructed to improve the stability and accuracy of support set prototypes using the cluster centers of query set samples.

The rest of this article is organized as follows. Section II describes the proposed MIEFSL method. The experimental results and analysis are presented in Section III. Finally, Section IV concludes this article.

II. Proposed Method

Fig. 1 illustrates the framework of the proposed MIEFSL algorithm. It consists of two stages to train the network for mining the rich spatial and spectral information of natural images and HSIs, respectively. In the first stage, the network is pretrained using natural images to learn spatial feature extractors. In the second stage, the spatial feature extractor is further fine-tuned using the HSIs to mitigate the interdomain differences between different types of images and to further learn domain-specific knowledge.

For pretraining with natural images, a random masking strategy is used to expand the support set and generate a masked support set. Considering the original and masked support sets as contrastive groups, we then design an MIE module to maximize the MI between support set samples with corresponding masked samples to improve the generalization ability of the pretrained model. In addition, an IPR module is proposed to learn stable category prototypes and also improve the network's discriminative ability.

For retraining with HSIs, an adaptive mapping layer is first used to reduce the dimension of HSI to 3. Then, the pretrained spatial feature extraction network and a deep 1-D-ResNet are used to extract spatial and spectral features, respectively. To mitigate the distribution differences between natural and HSIs, an IDA is performed on spatial features of the query set from source and target domains.

For testing, the labeled and unlabeled samples in the target dataset are simultaneously input into the VGG16 and deep 1-D-ResNet to extract the spatial and spectral features. Then, the extracted spatial and spectral features are concatenated and fed into a 1-nearest neighbor (NN) classifier for classification.

A. Data Processing

Assume that the source-domain natural image dataset D_s has C_s classes and the target-domain HSI dataset D_t has C_t classes, $C_s \geq C_t$ and $D_s \cap D_t = \emptyset$. For FSL, all samples in the source domain are labeled, while only very few labeled samples in the target domain are available (e.g., five labeled samples per class in the target domain are used in this article). In the source domain, 600 labeled samples per class are randomly selected for metatraining. In the target domain, the data augmentation operation of adding Gaussian noise is employed to expand the number of labeled samples from 5 to 200.

The construction of the support set and query set for both the source and target domains follows the same procedure. Taking the source dataset as an example, we use the N-way K-shot setting for episodic training. This is, in each metatask, we first randomly select N categories and then take K and K_q samples from each category to construct the support set and the query set, respectively. Denote the support and query sets as $S_s = \{(\boldsymbol{x}_i^{\mathrm{ss}}, y_i^{\mathrm{ss}})\}_{i=1}^{N\times K}$ and $Q_s = \{(\boldsymbol{x}_j^{\mathrm{sq}}, y_j^{\mathrm{sq}})\}_{j=1}^{N\times K_q}$. In this article, $N = C_t$, $K = 1$, and $K_q = 19$. The notations used are shown in Nomenclature.

B. Spatial and Spectral Feature Extraction

In order to capture rich spatial structure and texture information from natural images, we use the first seven layers of VGG16 as the spatial feature extraction network [28]. To extract the spatial features of HSI, an adaptive mapping layer is first used to reduce the dimension of HSI to 3. In detail, the adaptive mapping layer is a simple convolution operation via three convolution kernels of size $1\times 1\times \mathrm{ch}$, where ch is the input dimension of HSI. Then, the pretrained spatial feature extraction network can be used to extract HSI spatial features.

To extract spectral features of HSI, a deep 1-D convolutional residual network (1-D-ResNet) is designed, which consists of two 1-D convolutional layers, a spectral residual block, and a full connection (FC) layer.

C. Intradomain Prototype Rectification

In the traditional prototypical network, prototypes are obtained by averaging the features of support set samples from each class. The category prototype for the cth category in the support set of the source domain is computed as

$$\boldsymbol{A}_c^s = \frac{1}{K}\sum_{\boldsymbol{x}_i^{\mathrm{ss}}\in S_s^c} f_\theta\left(\boldsymbol{x}_i^{\mathrm{ss}}\right), \quad c = 1, \ldots, N \tag{1}$$

where S_s^c denotes the set of samples belonging to the cth category in the support set of the source domain. When the number of labeled samples in each category is limited (e.g., $K = 1$), the computed prototypes may be a deviation from the actual prototypes. This can be seen from the intradomain bias as

$$D_{\mathrm{intradomain}} = \mathbb{E}_{x_s'\sim P_c^s}\left[x_s'\right] - \mathbb{E}_{x_s\sim p_c^s}[x_s] \tag{2}$$

where P_c^s and p_c^s are the distributions of all samples and labeled samples of the cth category. The less information the

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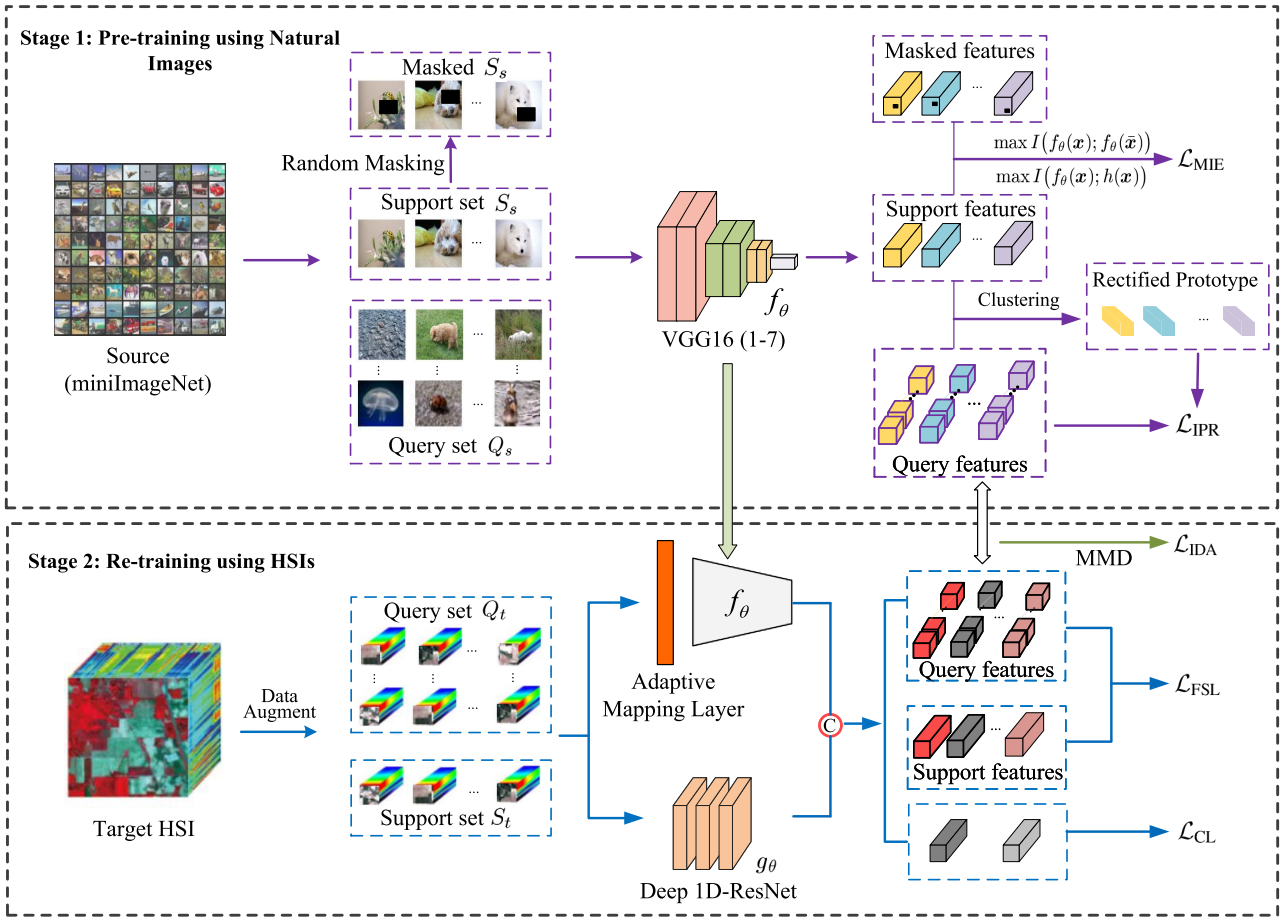


Fig. 1. Framework of the proposed MIEFSL method. It consists of two stages, i.e., pretraining on natural images and retraining on HSIs. In the pretraining stage, MIE and IPR modules are designed to improve the network's spatial feature extraction ability. In the retraining stage, an IDA module is used to align features of natural images and HSI, and FSL and CL are used to improve the discriminative ability.

distribution p_c^s contains, the more significant the intradomain bias $D_{\text{intradomain}}$. To diminish the intradomain bias, we adopt a distribution expansion strategy by increasing the available information about the actual distribution p_c^s.

In order to expand the actual distribution p_c^s, we use the K-means clustering algorithm to separate the support and query set samples into N clusters as $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \ldots, \boldsymbol{\mu}_N$. Then, the cluster centers are used to calibrate the category prototypes as

$$\widehat{\boldsymbol{A}}_c^s = \frac{w_c \boldsymbol{\mu}_c + \sum_{\boldsymbol{x}_i^{\text{ss}} \in S_s^c} f_\theta(\boldsymbol{x}_i^{\text{ss}})}{1+K} \tag{3}$$

where the weight w_c measures the similarity between the cluster centers and original prototypes as

$$w_c = \frac{\exp\left(\cos\left(\boldsymbol{\mu}_c, \boldsymbol{A}_c^s\right)\right)}{\sum_{c=1}^{N} \exp\left(\cos\left(\boldsymbol{\mu}_c, \boldsymbol{A}_c^s\right)\right)}. \tag{4}$$

Compared to the original prototype $\boldsymbol{A}_c^s$, the rectified prototype $\widehat{\boldsymbol{A}}_c^s$ is closer to the expected prototype.

For a query set sample $\boldsymbol{x}_j^{\text{sq}} \in Q_s$ with true label c_j, the network can predict its probability as

$$p\left(y_j = c_j | \boldsymbol{x}_j^{\text{sq}}\right) = \frac{\exp\left(-d\left(f_\theta(\boldsymbol{x}_j^{\text{sq}}), \widehat{\boldsymbol{A}}_j^s\right)\right)}{\sum_{j=1}^{N} \exp\left(-d\left(f_\theta\left(\boldsymbol{x}_j^{\text{sq}}\right), \widehat{\boldsymbol{A}}_j^s\right)\right)} \tag{5}$$

where $d(\cdot)$ represents the Euclidean distance.

The IPR loss for each episode is

$$\mathcal{L}_{\text{IPR}} = -\frac{1}{N \times K_q} \sum_{j=1}^{N \times K_q} \log p\left(y_j = c_j | \boldsymbol{x}_j^{\text{sq}}\right). \tag{6}$$

D. Mutual Information Enhancement

In order to further improve the generalization ability of the pretrained network and preserve the raw data information as much as possible, we design an MIE module for the source support set and its masked set.

In detail, for each sample $\boldsymbol{x}$ in the support set, 40% pixels of $\boldsymbol{x}$ are randomly selected for masking zero to obtain the corresponding new sample $\bar{\boldsymbol{x}}$. In the traditional CL, the network minimizes the cross-entropy loss between contrastive groups, which is equivalent to maximizing their MI [34]

$$\max I\left(f_\theta(\boldsymbol{x}); f_\theta(\bar{\boldsymbol{x}})\right). \tag{7}$$

CL only considers the shared information between two views and ignores the nonshared information. As shown in Fig. 2, maximizing the MI means enlarging the shared information, as shown in blue color. However, only maximizing the MI is not enough because some latent unseen sample information is not present in the shared part of the two views, and the

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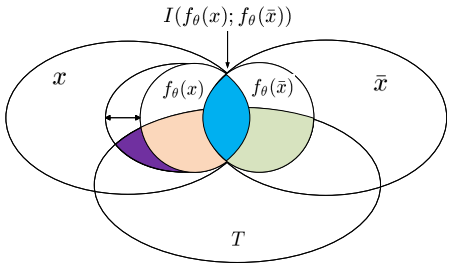


Fig. 2. Information diagrams in CL. Traditional CL only maximizes the shared information $I(f_\theta(x); f_\theta(\bar{x}))$. For the classification, some nonshared information (e.g., the orange and green parts) may be also important. Therefore, we aim to increase the classification-related nonshared information, e.g., enlarging the orange part to the purple region.

nonshared information (e.g., the orange and green parts) may be useful for the final classification task T [29]. To enhance nonshared classification-related information, it should increase $I(f_\theta(\boldsymbol{x}); T)$. As the task T is unknown during feature learning, we can alternatively increase $I(f_\theta(\boldsymbol{x}); \boldsymbol{x})$.

However, as the distribution of the original input $\boldsymbol{x}$ and the feature $f_\theta(\boldsymbol{x})$ is quite different, it is very difficult to directly maximize $I(f_\theta(\boldsymbol{x}); \boldsymbol{x})$ by taking $\boldsymbol{x}$ and $f_\theta(\boldsymbol{x})$ as the input of a neural network [29]. In the implementation, we utilize the feature of neural network intermediate layers $h(\boldsymbol{x})$ and maximize $I(f_\theta(\boldsymbol{x}); h(\boldsymbol{x}))$. If we consider a deep neural network as a Markov chain, we can obtain the corresponding data processing inequality [35]

$$I(Y; \boldsymbol{x}) \geq I(Y; h_{d_1}(\boldsymbol{x})) \geq I(Y; h_{d_2}(\boldsymbol{x})) \geq I(Y; f_\theta(\boldsymbol{x})) \quad (8)$$

where Y represents the real label information of input $\boldsymbol{x}$ and $h_{d_1}(\boldsymbol{x})$ and $h_{d_2}(\boldsymbol{x})$ represent the intermediate layer outputs at d_1th and d_2th layers of the network ($d_1 \leq d_2$), respectively. From (8), we can see that the MI of the intermediate layer can be considered as an approximation to the MI of the original input. In addition, the difference between intermediate layer feature $h(\boldsymbol{x})$ and final feature $f_\theta(\boldsymbol{x})$ is relatively small.

The intermediate layer output feature can be represented as [29]

$$h(\boldsymbol{x}) = h_d(\boldsymbol{x}) + \epsilon\sigma \quad (9)$$

where $\sigma = 0.1$ and $\varepsilon \sim \mathcal{N}(0, I)$. $h_d(\boldsymbol{x})$ is the output of the last maxpooling layer in the VGG16(1-7) network.

Finally, the optimization objective (7) can be changed to simultaneously extract the shared and nonshared classification-related information as follows:

$$\max I(f_\theta(\boldsymbol{x}); f_\theta(\bar{\boldsymbol{x}})) + I(f_\theta(\boldsymbol{x}); h(\boldsymbol{x})) + I(f_\theta(\bar{\boldsymbol{x}}); h(\bar{\boldsymbol{x}})). \quad (10)$$

The loss function corresponding to (10) can be written as

$$\mathcal{L}_{\boldsymbol{x}_i, \bar{\boldsymbol{x}}_i} = \ell(f_\theta(\boldsymbol{x}_i); f_\theta(\bar{\boldsymbol{x}}_i)) + \ell(f_\theta(\boldsymbol{x}_i); h(\boldsymbol{x}_i)) + \ell(f_\theta(\bar{\boldsymbol{x}}_i); h(\bar{\boldsymbol{x}}_i)) \quad (11)$$

where ℓ is a noise contrast estimation (NCE)-based loss function [29].

In the N-way one-shot case, there are $2N$ samples in the support set and masked support set. The loss for the MIE module is

$$\mathcal{L}_{\text{MIE}} = \frac{1}{2N} \sum_{i=1}^{N} \left(\mathcal{L}_{\boldsymbol{x}_i, \bar{\boldsymbol{x}}_i} + \mathcal{L}_{\bar{\boldsymbol{x}}_i, \boldsymbol{x}_i} \right). \quad (12)$$

E. Interdomain Distribution Alignment

In order to mitigate the distribution differences between natural images and HSIs, we perform IDA between the query set of source and target domains through the maximum mean difference (MMD) strategy

$$\mathcal{L}_{\text{IDA}} = \frac{1}{M} \left\| \sum_{\boldsymbol{x}^{\text{sq}} \in Q_s} \phi(f_\theta(\boldsymbol{x}^{\text{sq}})) - \sum_{\boldsymbol{x}^{\text{tq}} \in Q_t} \phi(f_\theta(\boldsymbol{x}^{\text{tq}})) \right\|_{\mathcal{H}}^2 \quad (13)$$

where $M = N \times K_q$ and $\phi(\cdot)$ is a nonlinear feature map that induces a reproducing kernel Hilbert space $\mathscr{H}$.

In the pretraining stage, the total loss is defined as

$$\mathcal{L}_s^{\text{total}} = \mathcal{L}_{\text{IPR}} + \mathcal{L}_{\text{MIE}} + \mathcal{L}_{\text{IDA}}. \quad (14)$$

F. Fine-Tuning on Target Domain

On the target domain, the VGG16-based spatial feature extractor f_θ and the 1-D-ResNet-based spectral feature extractor g_θ are used to extract spatial and spectral features of HSI, respectively. Then, the extracted features are concatenated and denoted as follows: $\overline{f}_\theta(\boldsymbol{x}) = [f_\theta(\boldsymbol{x}); g_\theta(\boldsymbol{x})]$.

Denote the support and query sets of target domains as $S_t = \{(\boldsymbol{x}_i^{\text{ts}}, y_i^{\text{ts}})\}_{i=1}^{N \times K}$ and $Q_t = \{(\boldsymbol{x}_j^{\text{tq}}, y_j^{\text{tq}})\}_{j=1}^{N \times K_q}$, respectively. Similar to the operations on the source domain, we can calculate the target category prototypes and class membership probability according to (1) and (5), respectively. Consequently, the FSL loss on the target query set $\mathcal{L}_{\text{FSL}}$ can be defined as

$$\mathcal{L}_{\text{FSL}} = -\frac{1}{N \times K_q} \sum_{j=1}^{N \times K_q} \log p\left(y_j^{\text{tq}} = c_j | \boldsymbol{x}_j^{\text{tq}}\right). \quad (15)$$

To further improve the discriminative performance, the CL strategy is used for training the model. Specifically, in a batch of size B, for each feature $\overline{f}_\theta(\boldsymbol{x}_j)$, we perform two different random dropout operations on the feature to generate two different attempts $\overline{f}_\theta(\boldsymbol{x}_j^1)$ and $\overline{f}_\theta(\boldsymbol{x}_j^2)$ and can calculate the contrastive loss between them as

$$\mathcal{L}_{\text{CL}} = \frac{1}{B} \sum_{j=1}^{B} \left(\ell\left(\overline{f}_\theta(\boldsymbol{x}_j^1), \overline{f}_\theta(\boldsymbol{x}_j^2)\right) + \ell\left(\overline{f}_\theta(\boldsymbol{x}_j^2), \overline{f}_\theta(\boldsymbol{x}_j^1)\right) \right). \quad (16)$$

The total loss for the model can be expressed as

$$\mathcal{L}_t^{\text{total}} = \mathcal{L}_{\text{FSL}} + \mathcal{L}_{\text{CL}}. \quad (17)$$

III. Experimental Results and Analysis

A. Datasets

In the experiments, we adopt the miniImageNet natural image dataset as the source domain D_s and use the Indian Pines (IP), University of Pavia (UP), and Salinas (SA) hyperspectral datasets as the target domain D_t, respectively. Table I provides the detailed information about the target-domain datasets.

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Algorithm 1 MIEFSL Training Process

Input: The feature extractor f_θ
The support set S_s and query set Q_s in D_s
The support set S_t and query set Q_t in D_t
Output: The loss of model
if episode $<$ 1000 **then**
  Train with S_s and Q_s
  Calculate the prototypes by Eq. (1)
  for $\boldsymbol{x}_i^{ss} \in S_s$, $\boldsymbol{x}_j^{sq} \in Q_s$ **do**
    Rectify the prototypes by Eq. (3)
    Calculate $\mathcal{L}_{\text{IPR}}$ by Eq. (6)
  end for
  for $\boldsymbol{x}_i^{ss} \in S_s$ **do**
    Calculate $\mathcal{L}_{\text{MIE}}$ by Eq.(12)
  end for
  for $\boldsymbol{x}^{sq} \in Q_s$, $\boldsymbol{x}^{tq} \in Q_t$ **do**
    Calculate $\mathcal{L}_{\text{IDA}}$ by Eq. (13)
  end for
  Return the $\mathcal{L}_s^{\text{total}}$ of an episode by Eq. (14)
end if
Transfer the pre-training network weights
if episode $<$ 1000 **then**
  Train with S_t and Q_t
  for $\boldsymbol{x}_i^{ts} \in S_t$, $\boldsymbol{x}_j^{tq} \in Q_t$ **do**
    Calculate $\mathcal{L}_{\text{FSL}}$ by Eq. (15)
    Calculate $\mathcal{L}_{\text{CL}}$ by Eq. (16)
  end for
  Return the $\mathcal{L}_t^{\text{total}}$ of an episode by Eq. (17)
end if

TABLE I
DETAILED INFORMATION ABOUT THREE TARGET-DOMAIN DATASETS

Dataset	Indian Pines	University of Pavia	Salinas
Location	Indiana, USA	Pavia, Italy	California, USA
Sensor Name	AVIRIS	ROSIS-03	AVIRIS
Image Size	145×145	610×340	512×217
Number of Bands	224	115	224
Number of Retained Bands	200	103	204
Spatial Resolution	20m	1.3m	3.7m
Wavelength Range	$0.4 \sim 2.5\mu m$	$0.43 \sim 0.86\mu m$	$0.4 \sim 2.5\mu m$

1) Source-Domain Dataset: The miniImageNet dataset is widely used in the fields of metalearning and FSL. It is a subset of the ImageNet dataset. From the ImageNet1k, 100 categories are randomly selected. The training set consists of 600 labeled images per class. Each image has the size of 84 × 84.

2) Target-Domain Dataset:

a) Indian Pines: It was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) in 1992 over a stand of Indian Pine Trees in Indiana, USA. The IP data have 200 spectral bands ranging from 400 to 2500 nm. The image has a size of 145 × 145 pixels. It contains 16 categories and 10 249 labeled pixels.

b) University of Pavia: The UP dataset was captured by the Reflective Optics Spectrographic Imaging System (ROSIS-03) over the city of Pavia, Italy, in 2003. The original data have 115 spectral bands within the wavelength range of 0.43–0.86 μm, with a spatial resolution of 1.3 m. By deleting 12 noisy bands, the remaining 103 spectral bands are used for classification. The image size is 610 × 340. The UP data contain nine categories and 42 776 labeled pixels.

c) Salinas: The SA dataset was captured by the AVIRIS over the Salinas Valley in California, USA. This data has a 3.7-m spatial resolution and 224 spectral bands. By deleting bad bands 108–112, 154–167, and 224, the remaining 204 bands are used for classification. The image size is 512 × 217. The SA data contain 16 categories and 54 129 labeled pixels. Tables II–IV show the land cover classes and number of samples for three target domain data sets, respectively. Figs. 3–5 provide the pseudo-color composite image and the corresponding ground-truth map of three data sets.



Fig. 3. Pseudocolor composite image and ground-truth map of IP.

TABLE II
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN IP

Class	Name	Number of samples
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-trees	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93
Total		10249

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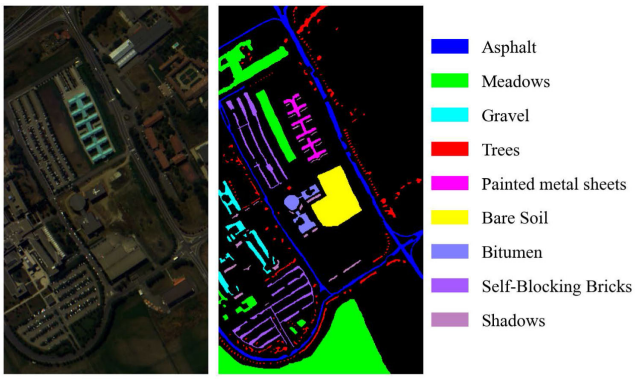


Fig. 4. Pseudocolor composite image and ground-truth map of UP.

TABLE III
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN UP

Class	Name	Number of samples
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Sheets	1345
6	Bare soil	5029
7	Bitumen	1330
8	Bricks	3682
9	Shadow	947
Total		42776

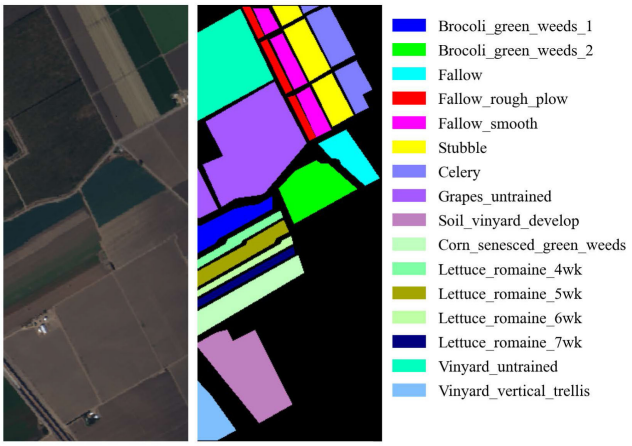


Fig. 5. Pseudocolor composite image and ground-truth map of SA.

B. Experimental Setting

To evaluate the performance of the proposed MIEFSL, we compare some classical HSI classification methods, such as HybridSN [14] and SSRN [13], and state-of-the-art FSL methods, including DCFSL [22], CMFSL [24], Gia-CFSL [23], RPCL-FSL [25], ADAFSL [26], and HFSL [27].

For all these methods, only five labeled samples per class in the target domain are used. For HybridSN and SSRN, five labeled samples per class in the target domain are used to train the model for directly classifying the unlabeled target samples. For FSL methods, all settings follow the corresponding original literature. For DCFSL, CMFSL, Gia-CFSL, RPCL-FSL, and ADAFSL, the number of labeled samples per class is 200 and 5 for the source and target domains, respectively. For HFSL and our MIEFSL, the source-domain dataset is the miniImageNet, in which 600 labeled samples per class are used for training.

TABLE IV
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN SA

Class	Name	Number of samples
1	Brocoli_green_weeds_1	2009
2	Brocoli_green_weeds_2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807
Total		54129

We adopt an episode training strategy, constructing N-way K-shot classification tasks on the support set, where N is set as the number of classes in the testing dataset and K is set to 1. In the query set, there are 19 labeled samples per class.

All experiments are implemented on the NVIDIA GeForce RTX2080Ti platform based on the PyTorch framework, with a computer memory of 64 GB. The Adam optimizer with a learning rate of 0.001 is used. The classification evaluation metrics are the overall accuracy (OA), average accuracy (AA), and κ coefficient. The experiments are randomly run ten times, and the averaged results are reported.

C. Experimental Results

Table V shows the classification results of different methods on the IP dataset. It can be seen that our MIEFSL method outperforms other methods in terms of OA, AA, and κ coefficient consistently. Specifically, compared with the classification methods, HybridSN and SSRN, the proposed MIEFSL improves the OA by at least 12%. This verifies the effectiveness of cross-domain FSL by pretraining on the source dataset and fine-tuning the target domain. Compared with the latest FSL methods, e.g., RPCL-FSL and ADAFSL, our MIEFSL outperforms the OA and κ coefficient by about 2% and 3%, respectively. Among the 16 ground-truth classes, our MIEFSL provides the best results in six classes. On the subclasses of "Soybean," i.e., Classes 10–12, MIEFSL shows the best results. On the subclasses of "Corn," i.e., Classes 2–4, MIEFSL also shows comparable

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TABLE V
CLASSIFICATION RESULTS (%) ON IP

Class	HybridSN [14]	SSRN [13]	DCFSL [22]	CMFSL [24]	Gia-CFSL [23]	RPCL-FSL [25]	ADAFSL [26]	HFSL [27]	MIEFSL
1	88.14	38.54	96.21	94.12	93.17	98.29	**100**	98.78	99.27
2	64.13	65.27	43.10	48.13	39.14	**66.11**	33.66	62.54	55.62
3	60.54	19.75	54.09	**83.12**	41.30	66.05	56.21	46.17	65.44
4	43.27	24.33	81.35	92.57	72.59	88.28	90.67	81.68	**93.07**
5	51.04	80.25	80.05	**94.85**	71.63	77.85	83.89	72,86	82.19
6	77.68	84.82	90.29	96.42	81.28	89.56	**99.31**	88.64	87.36
7	79.24	33.74	99.13	**100**	96.52	**100**	99.84	**100**	**100**
8	89.74	**100**	83.74	89.45	84.10	93.95	97.89	99.68	97.76
9	23.87	60.29	**100**	**100**	**100**	**100**	**100**	**100**	**100**
10	56.18	60.14	60.89	70.15	58.25	74.17	68.05	80.15	**83.33**
11	81.98	69.55	57.69	38.79	59.44	64.50	76.90	70.71	**82.97**
12	46.71	48.18	47.93	50.16	43.89	66.14	51.87	50.12	**92.15**
13	90.28	86.13	98.05	97.36	97.50	97.05	97.50	**98.50**	96.82
14	70.13	**95.21**	83.14	83.14	75.84	87.07	84.52	92.54	93.83
15	32.84	34.81	67.09	65.01	65.20	84.54	**98.69**	93.21	95.80
16	26.19	60.42	96.17	97.76	98.30	97.84	**100**	96.78	98.84
OA	65.73 ±3.16	61.75 ±4.24	65.28 ±2.29	67.02 ±1.34	63.28 ±3.74	76.29 ±2.28	75.04 ±1.15	75.18 ±3.26	**78.34 ±2.71**
AA	61.05 ±1.35	59.81 ±6.87	75.64 ±1.73	82.13 ±0.63	74.93 ±2.69	85.15 ±1.39	83.91 ±1.09	84.01 ±1.21	**86.45 ±1.72**
κ	64.89 ±2.76	56.97 ±7.63	58.79 ±2.81	63.29 ±1.24	58.89 ±3.46	72.49 ±2.56	71.85 ±0.87	72.33 ±2.63	**75.30 ±2.04**

TABLE VI
CLASSIFICATION RESULTS (%) ON UP

Class	HybridSN [14]	SSRN [13]	DCFSL [22]	CMFSL [24]	Gia-CFSL [23]	RPCL-FSL [25]	ADAFSL [26]	HFSL [27]	MIEFSL
1	59.71	93.12	80.56	72.35	72.56	84.35	88.13	76.89	**93.19**
2	79.23	**96.77**	82.18	91.28	82.18	82.41	85.95	86.57	91.81
3	53.19	53.31	62.32	**79.63**	62.32	70.11	77.84	77.48	75.41
4	86.27	67.12	92.81	90.22	92.81	93.16	**94.31**	88.46	87.09
5	**100**	99.58	98.43	99.87	98.43	99.33	99.18	99.96	99.93
6	91.10	61.52	75.68	76.11	71.58	81.28	82.44	**98.58**	84.89
7	98.04	73.34	80.63	95.38	80.63	84.41	83.47	**98.19**	88.49
8	43.66	81.39	65.91	75.73	65.91	85.37	78.12	**97.92**	81.86
9	98.01	96.57	97.15	98.46	98.85	98.62	94.67	99.58	**100**
OA	74.18 ±2.03	77.81 ±2.78	81.35 ±2.31	85.34 ±1.02	81.66 ±2.67	86.19 ±2.98	86.45 ±1.15	88.22 ±2.13	**90.31 ±1.62**
AA	88.48 ±1.26	78.35 ±3.24	81.87 ±1.29	86.46 ±0.97	80.58 ±3.12	85.13 ±3.57	85.50 ±1.32	**90.15 ±2.27**	88.74 ±1.79
κ	69.83 ±3.87	70.05 ±5.63	75.61 ±3.25	81.21 ±1.13	75.54 ±3.74	82.27 ±2.19	82.74 ±0.67	84.83 ±3.75	**88.01 ±3.58**

results with other methods. The effectiveness of MIEFSL in distinguishing subtle differences between similarity classes may be attributed to the prototype rectification and MIE, which improves the discriminative ability of the proposed method.

Table VI shows the classification results on the UP dataset. Our MIEFSL also provides the highest OA and κ values than other methods. Although the AA is slightly lower than HFSL, the OA of MIEFSL is 2% higher than HFSL. It should be noted that both HFSL and MIEFSL use the miniImageNet as the source dataset to train the FSL model and adopt the VGG16(1-7) as the spatial feature extraction subnetwork. The difference is that we generate a masked support set and use an MIE strategy to improve the discriminative and generalization ability of the network in the CL process. In addition, the rectified prototypes are likely to improve the final classification.

Table VII shows the classification results on the SA dataset. Our MIEFSL generates the highest OA, AA, and κ coefficient with values of 91.72%, 95.87%, and 90.70%, respectively. Compared with HFSL, MIEFSL improves the OA, AA, and κ by at least 2%. In the SA dataset, Classes 8 and 15 are spatially adjacent and have similar spectral characteristics, which are easily to be confused. Notwithstanding, our MIEFSL shows the best result in Class 15 and the third best result in Class 8.

Figs. 6–8 visually demonstrate the classification map of different methods on the three datasets. It can be seen that the classification maps produced by our method are closer to the ground truth. In particular, we have marked the region

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TABLE VII
CLASSIFICATION RESULTS (%) ON SA

Class	HybridSN [14]	SSRN [13]	DCFSL [22]	CMFSL [24]	Gia-CFSL [23]	RPCL-FSL [25]	ADAFSL [26]	HFSL [27]	MIEFSL
1	99.87	96.17	99.58	99.18	99.19	99.05	99.19	99.87	**100**
2	**100**	97.01	99.43	99.33	98.84	99.52	99.03	96.43	98.23
3	97.21	92.56	90.09	**98.23**	86.10	90.61	93.21	95.69	92.87
4	95.93	96.89	99.24	99.26	99.28	98.69	99.42	99.68	**99.71**
5	88.75	90.03	91.53	94.46	93.32	89.81	**96.18**	88.27	94.95
6	99.49	99.24	99.02	98.96	**100**	97.95	99.01	99.81	99.52
7	**100**	99.87	96.22	99.49	99.83	99.05	95.17	96.66	97.29
8	75.64	85.11	70.26	70.11	**91.52**	77.32	75.22	72.47	81.48
9	94.27	99.34	99.51	95.38	99.79	99.71	99.53	99.86	**99.97**
10	89.76	93.13	83.03	93.37	73.72	86.43	91.48	**94.47**	84.85
11	95.72	95.62	96.45	98.37	93.89	99.32	99.44	99.32	**99.72**
12	90.45	99.08	**99.63**	93.12	99.17	98.77	98.90	98.26	95.27
13	98.81	89.32	99.14	**100**	99.23	99.46	99.67	99.15	98.79
14	57.81	95.33	98.16	**99.85**	90.27	98.40	98.17	97.09	99.15
15	60.21	54.23	74.19	85.75	51.04	79.92	80.68	73.42	**85.78**
16	**100**	97.36	90.46	92.30	93.01	91.48	97.84	95.70	97.45
OA	84.39 ±1.24	85.14 ±2.17	87.62 ±2.79	89.18 ±1.62	88.51 ±1.76	90.18 ±1.92	90.23 ±1.37	89.23 ±2.65	**91.72 ±2.33**
AA	90.75 ±0.71	91.02 ±1.04	92.87 ±2.08	92.54 ±1.03	91.28 ±1.62	94.09 ±1.31	95.60 ±0.49	93.36 ±1.97	**95.87 ±1.48**
κ	81.69 ±1.35	83.97 ±1.78	86.26 ±3.07	87.39 ±2.23	87.14 ±1.25	89.09 ±2.10	90.18 ±1.04	88.19 ±2.68	**90.70 ±2.61**

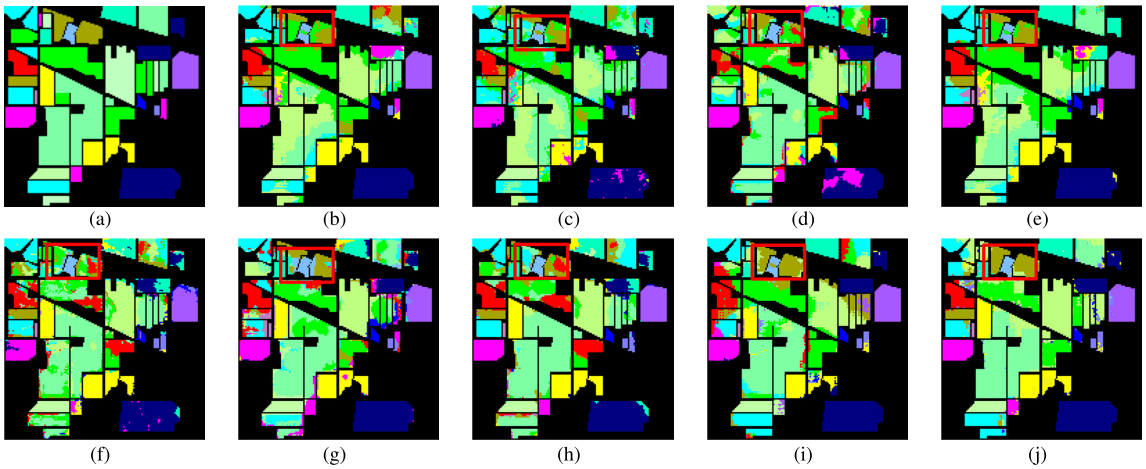


Fig. 6. Classification map of different methods on IP. (a) Ground truth. (b) HybridSN. (c) SSRN. (d) DCFSL. (e) CMFSL. (f) Gia-CFSL. (g) RPCL-FSL. (h) ADAFSL. (i) HFSL. (j) MIEFSL.

TABLE VIII
ABLATION ANALYSIS ON DIFFERENT MODULES

	Base	Base-IPR	Base-MIE	Base-IDA	MIEFSL
IP	73.76	75.21	76.73	76.12	78.34
UP	87.54	87.95	89.62	89.50	90.31
SA	86.51	87.84	90.13	89.01	91.72

TABLE IX
ABLATION ANALYSIS ON THE MIE MODULE

	Base	Base-CL	Base-MIE
IP	73.76	75.17	76.73
UP	87.54	88.53	89.62
SA	86.51	89.46	90.13

where our MIEFSL clearly outperforms other methods in red rectangles.

D. Ablation Analysis

The proposed MIEFSL mainly consists of three modules: IPR, MIE, and IDA. To validate the effectiveness of different modules, we have performed ablation experiments on three datasets in Table VIII.

We consider four variants in the ablation analyses.

1) *Base:* The general FSL framework without three modules.

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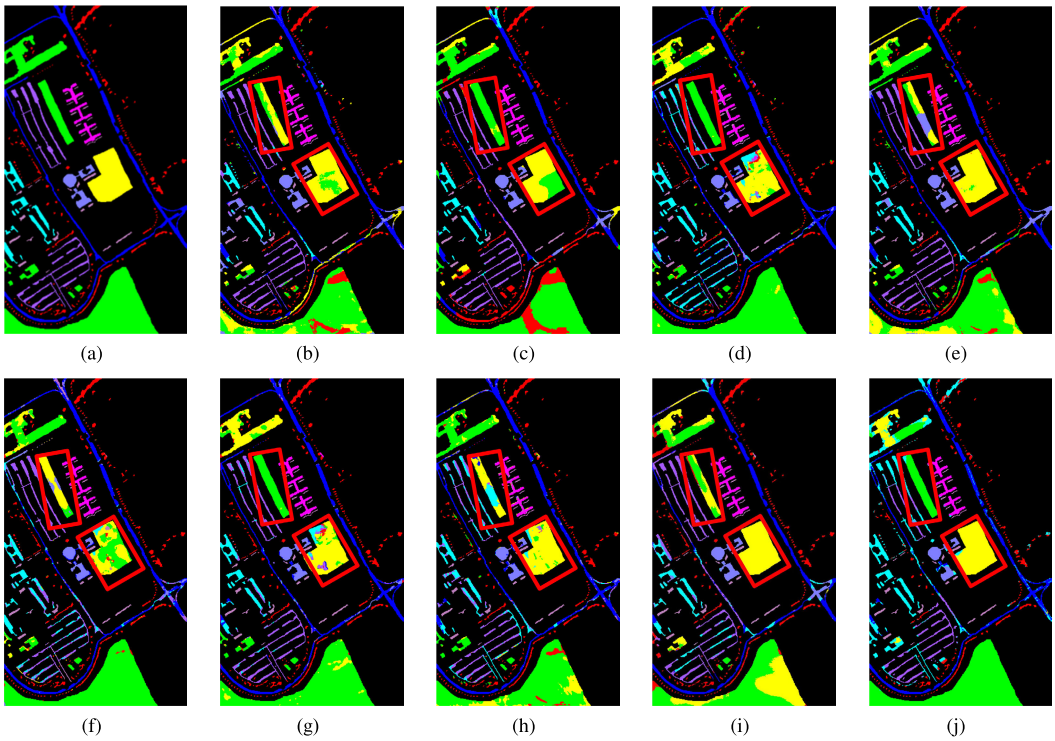


Fig. 7. Classification map of different methods on UP. (a) Ground truth. (b) HybridSN. (c) SSRN. (d) DCFSL. (e) CMFSL. (f) Gia-CFSL. (g) RPCL-FSL. (h) ADAFSL. (i) HFSL. (j) MIEFSL.

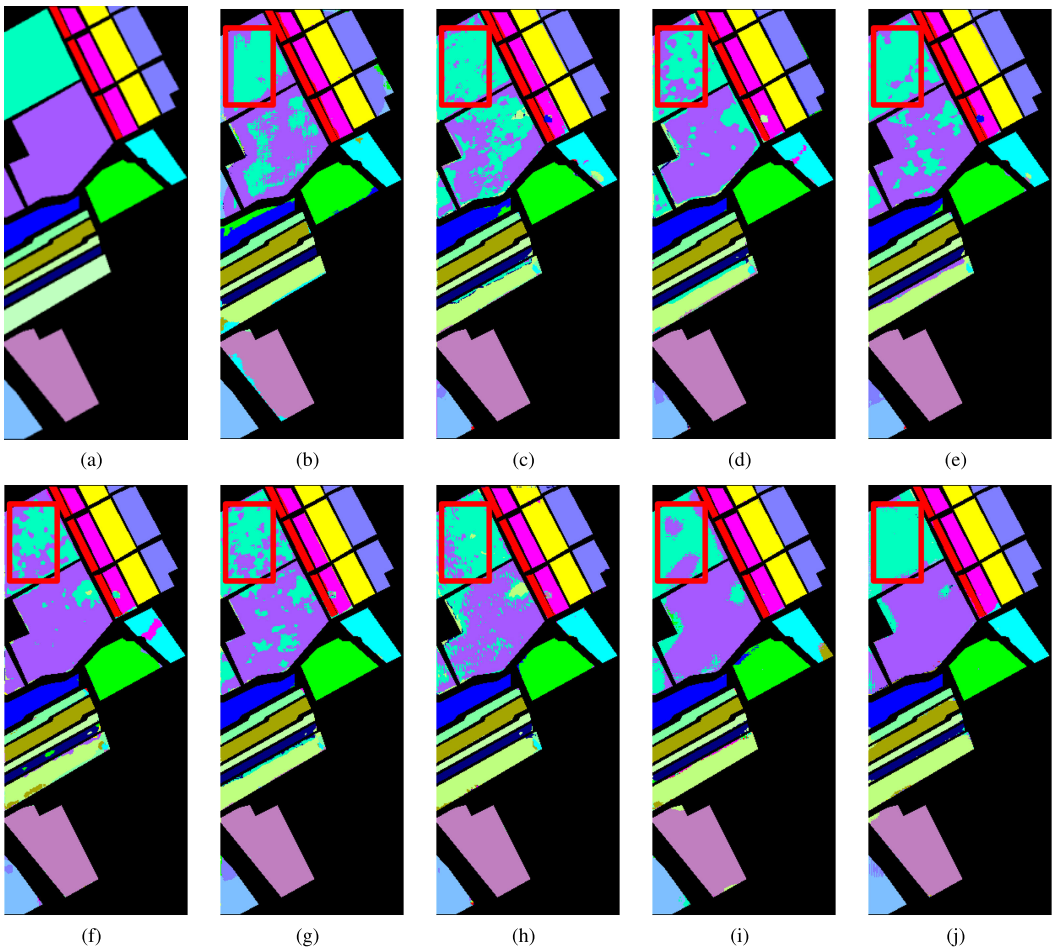
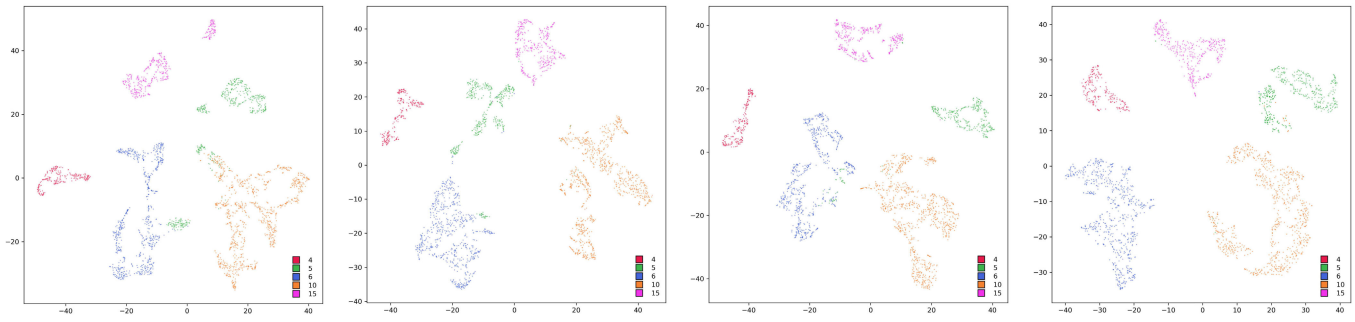


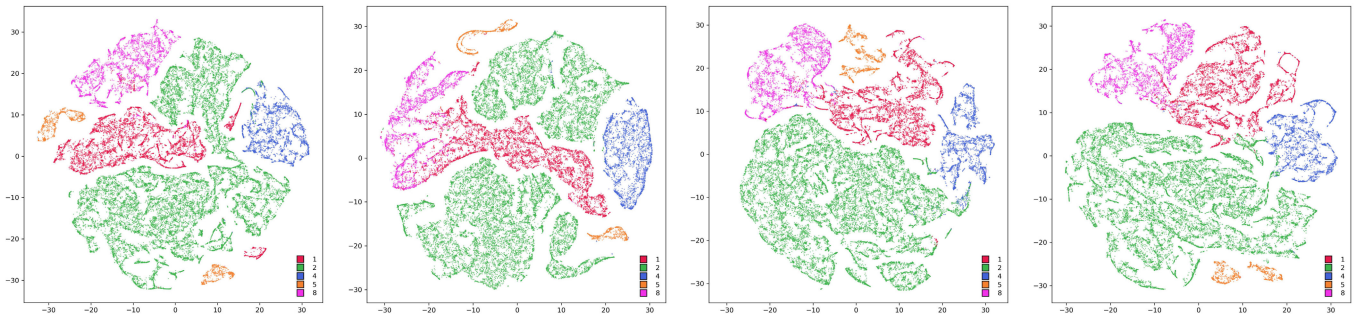
Fig. 8. Classification map of different methods on SA. (a) Ground truth. (b) HybridSN. (c) SSRN. (d) DCFSL. (e) CMFSL. (f) Gia-CFSL. (g) RPCL-FSL. (h) ADAFSL. (i) HFSL. (j) MIEFSL.

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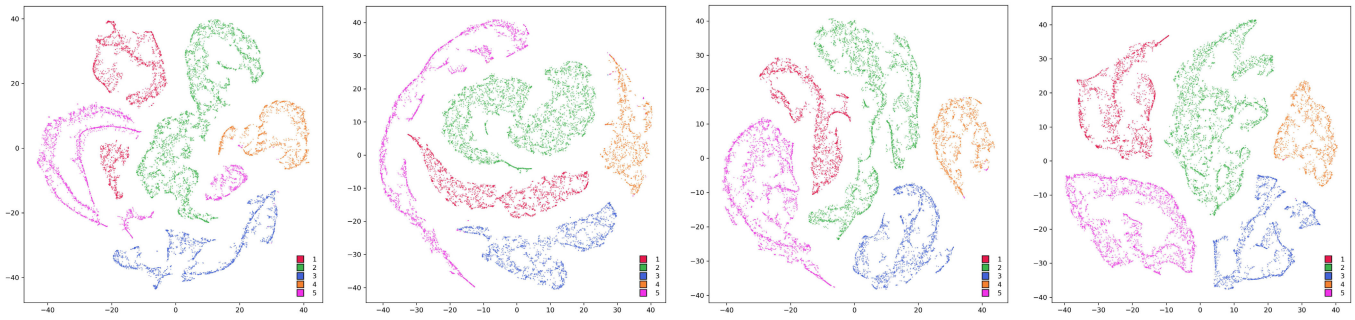
(a) (b) (c) (d)

Fig. 9. Feature visualization of different modules using the t-SNE on IP. (a) Adding IDA module to the general FSL framework (Base-IDA). (b) Adding IDA module and IPR module to the general FSL framework (Base-IDA-IPR). (c) Adding IDA module and MIE module to the general FSL framework (Base-IDA-MIE). (d) Adding all three modules to the general FSL framework (MIEFSL).



(a) (b) (c) (d)

Fig. 10. Feature visualization of different modules using the t-SNE on UP. (a) Adding IDA module to the general FSL framework (Base-IDA). (b) Adding IDA module and IPR module to the general FSL framework (Base-IDA-IPR). (c) Adding IDA module and MIE module to the general FSL framework (Base-IDA-MIE). (d) Adding all three modules to the general FSL framework (MIEFSL).



(a) (b) (c) (d)

Fig. 11. Feature visualization of different modules using the t-SNE on SA. (a) Adding IDA module to the general FSL framework (Base-IDA). (b) Adding IDA module and IPR module to the general FSL framework (Base-IDA-IPR). (c) Adding IDA module and MIE module to the general FSL framework (Base-IDA-MIE). (d) Adding all three modules to the general FSL framework (MIEFSL).

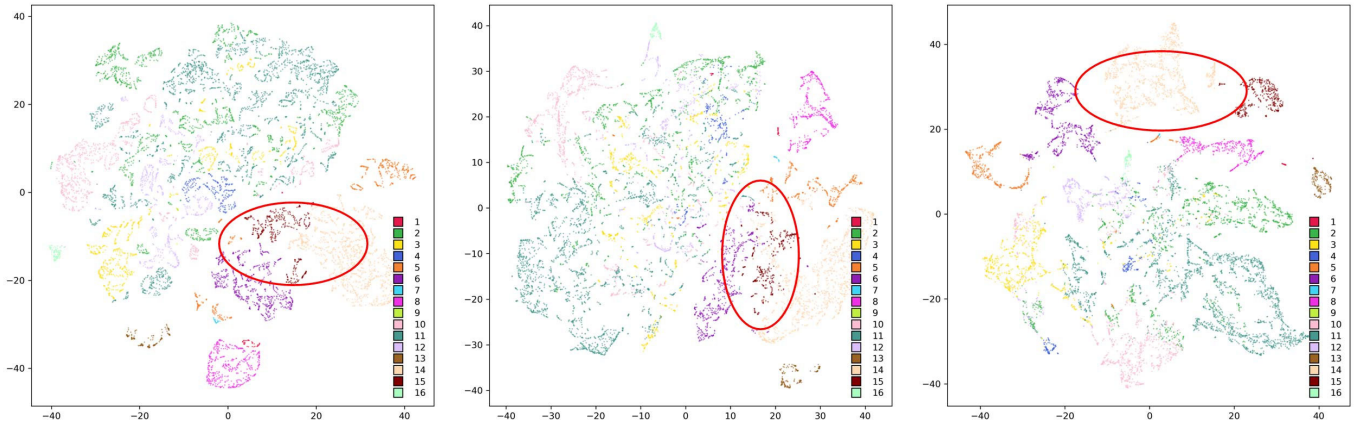
2) *Base-IPR:* Only adding the IPR module into the base model.
3) *Base-MIE:* Only adding MIE module into the base model.
4) *Base-IDA:* Only adding IDA module into the base model.

The OAs of different variants are listed in Table VIII. It can be observed that each module improves the base model, and the incorporation of all three modules produces the best results.

Among the three modules, the MIE module shows the best results. From (10), we can see that the MIE module contains three terms where the first term corresponds to the original CL to maximize the shared information and the last two terms are enhancement parts to extract the nonshared classification-related information. We further analyze the effect of different terms in Table IX, where "Base-CL" denotes the base model with only the CL term. It can be seen that enhancing the MI improves the OA by 1.56%, 1.09%, and 0.67% on three datasets, respectively.

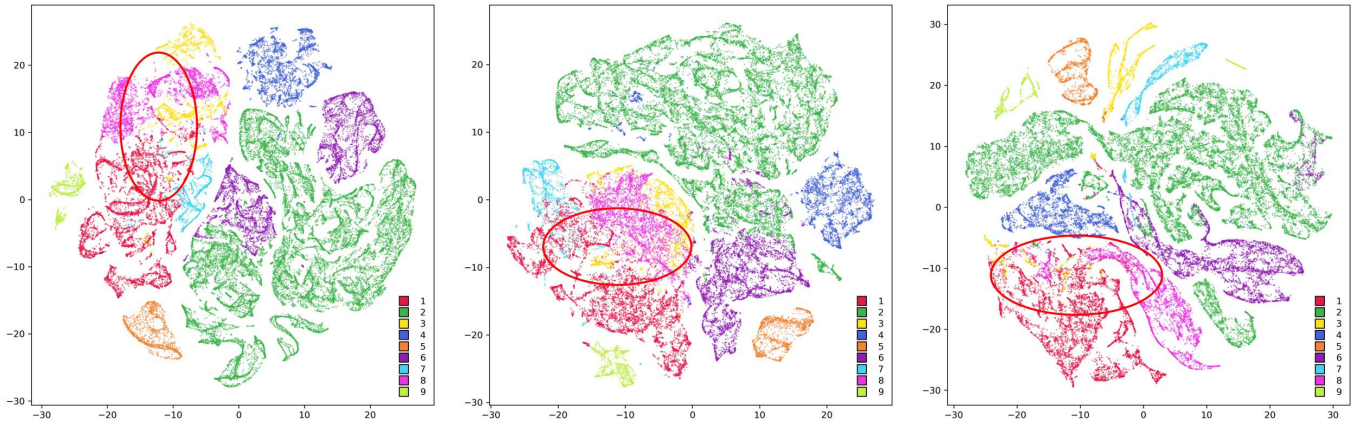
To visually assess the effectiveness of different modules, we select five categories from each dataset and utilize the t-distributed stochastic neighbor embedding (t-SNE) strategy to visualize the extracted features of Base-IDA, Base-IDA-IPR, Base-IDA-MIE, and MIEFSL in Figs. 9–11. It can be seen that either the IPR or MIE module can improve

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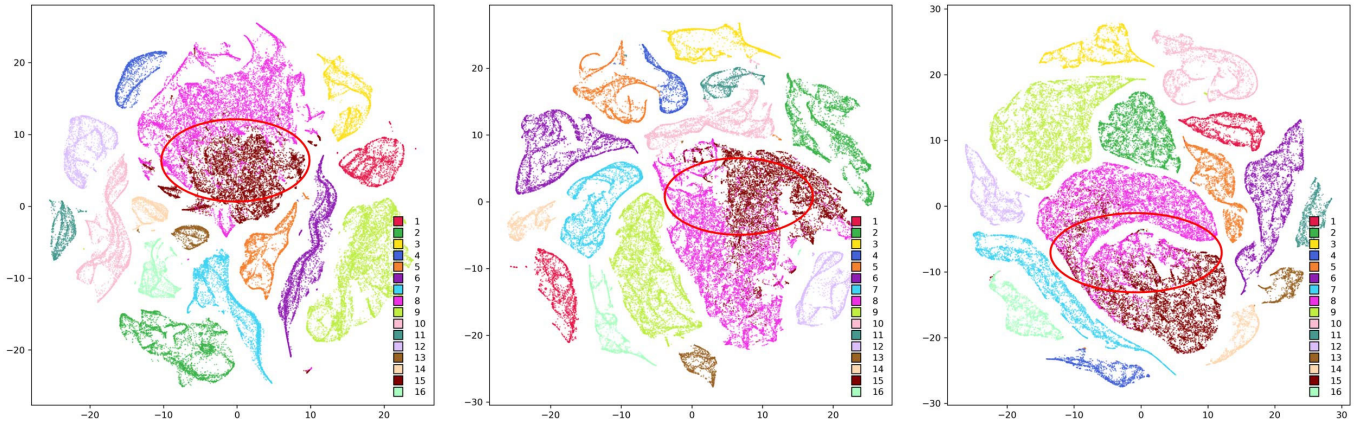
(a) (b) (c)

Fig. 12. Feature visualization of different methods using the t-SNE on IP. (a) DCFSL. (b) Gia-CFSL. (c) MIEFSL.



(a) (b) (c)

Fig. 13. Feature visualization of different methods using the t-SNE on UP. (a) DCFSL. (b) Gia-CFSL. (c) MIEFSL.



(a) (b) (c)

Fig. 14. Feature visualization of different methods using the t-SNE on SA. (a) DCFSL. (b) Gia-CFSL. (c) MIEFSL.

intraclass compactness. By combining all these submodules, our proposed MIEFSL shows good discriminative performance with small intraclass scatter and large interclass separation.

E. Feature Visualization

To visually assess the feature extraction abilities of different methods, we utilize the t-SNE strategy to visualize the extracted features of DCFSL, Gia-CFSL, and our MIEFSL on the three datasets in Figs. 12–14. It can be observed that the features of the same class extracted by our proposed method are relatively compact, while those of different classes are relatively scattered.

F. Analysis of Computational Complexity

Table X shows the computational complexity of different algorithms, mainly using four metrics: training time, testing time, floating-point operations (FLOPs), and parameters.

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TABLE X
COMPUTATIONAL TIME (S), FLOPS (M), AND PARAMETERS (M) OF DIFFERENT METHODS ON THREE DATASETS

Dataset	Metric	Methods								
		HybridSN	SSRN	DCFSL	CMFSL	Gia-CFSL	RPCL-FSL	ADAFSL	HFSL	MIEFSL
IP	Training time	9.31	46.95	1907.86	2563.25	4534.13	732.18	2013.72	1276.09	1855.84
	Testing time	3.42	3.78	1.93	4.75	1.96	3.64	4.05	17.92	12.31
	FLOPS	253.64	162.21	48.54	55.61	26837.44	48.53	43.84	408.98	387.46
	#Params	5.12	0.36	4.26	1.22	0.85	4.27	4.25	6.27	5.23
UP	Training time	5.48	13.55	1297.53	1869.28	2721.44	646.42	1836.08	621.56	1397.49
	Testing time	14.63	9.56	7.06	6.82	7.39	16.35	17.31	27.69	13.36
	FLOPS	254.64	83.15	47.72	28.67	26821.54	47.72	43.02	408.16	387.02
	#Params	5.12	0.21	4.25	0.63	0.84	4.26	4.24	6.13	5.12
SA	Training time	9.31	48.25	1991.98	2679.24	4612.65	1009.15	2155.73	1519.49	1969.73
	Testing time	18.48	20.03	10.62	8.74	10.81	22.79	22.93	72.06	28.19
	FLOPS	254.64	165.47	48.54	57.24	26839.67	48.54	43.84	409.09	388.57
	#Params	5.12	0.37	4.26	1.25	0.85	4.27	4.25	6.28	5.23

It can be seen that traditional deep learning algorithms (i.e., HybridSN and SSRN) are trained solely on the target domain, which is much faster than FSL methods with training on both source and target domains. However, traditional deep learning algorithms usually show worse results than FSL methods due to the limited number of labeled samples. Compared with some classic FSL algorithms (e.g., CMFSL, Gia-CFSL, and ADAFSL), our MIEFSL provides a shorter training time, largely due to its more streamlined network structure and fewer training episodes. However, it has a longer training time compared to some new FSL algorithms (i.e., RPCL-FSL and HFSL) because it involves more time on training on natural image datasets and computing MI. Overall, our proposed MIEFSL achieves better classification performance with an acceptable computational cost.

IV. CONCLUSION

In this article, a novel method called MIEFSL has been proposed to solve the FSL classification of HSIs. Considering that natural images have rich spatial texture information, MIEFSL uses natural image datasets as a source domain to transfer metaknowledge from natural images to HSIs. To solve the prototype instability problem, an IPR module is proposed to learn stable category prototypes through a distribution expansion strategy and also to improve the network's discriminative ability. In order to further improve the generalization performance of the network, we design an MIE module to maximize the MI between support set samples with corresponding masked samples. In addition, an IDA module is designed to mitigate the distribution differences between natural images and HSIs. The effectiveness of the proposed MIEFSL has been verified by experiments on three public HSI datasets. Compared with existing methods, the OAs of MIEFSL on IP, UP, and SA datasets are increased 2.05%∼16.59%, 2.09%∼16.13%, and 1.49%∼7.33%, respectively. In the future, we will incorporate textual information (e.g., label text) into the FSL framework. By fusing textual modality information such as label annotations or geographic descriptions with visual modality information, it is likely to improve the classification performance on ground objects that are difficult to be classified.

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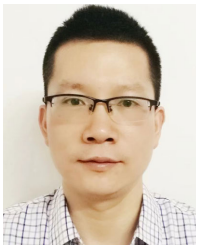
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